

Gender, Attention-Deficit/Hyperactivity Disorder, and Reading Disability in a Population-Based Birth Cohort



WHAT'S KNOWN ON THIS SUBJECT: Results of previous studies have shown a strong association between ADHD and RD in clinical samples. However, studies on community samples have been scarce, and the comorbidity rates reported have not been consistent.



WHAT THIS STUDY ADDS: In a population-based birth cohort, approximately half of the children with ADHD had comorbid RD (both boys and girls). The risk of RD associated with ADHD was significantly higher in girls than in boys.

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KEY WORDS

ADD, ADHD, dyslexia, reading disability, comorbidity, epidemiology

ABBREVIATIONS

ADHD—attention-deficit/hyperactivity disorder

RD—reading disability

HR—hazard ratio

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abstract



OBJECTIVE: To determine the incidence of reading disability (RD) among children with and without research-identified attention-deficit/hyperactivity disorder (ADHD), separately according to gender, in a population-based birth cohort.

METHOD: Subjects included all children born in 1976–1982 remaining in Rochester, Minnesota, after 5 years of age ($n = 5718$). Information from medical, school, and private tutorial records was abstracted. Cumulative incidence of RD, by any of 3 RD formulas, in children with and without ADHD and corresponding hazard ratios (HRs) were calculated separately according to gender.

RESULTS: Cumulative incidence of RD by the age of 19 years was significantly higher in children with ADHD (51% in boys, 46.7% in girls) compared with those without ADHD (14.5% in boys, 7.7% in girls). Among children with ADHD, the risk for RD was similar in boys versus girls (HR: 1.0). However, among children without ADHD, boys were 2.0 times more likely than girls to meet RD criteria. Among girls, the HR for the risk for RD associated with ADHD (versus those without ADHD) was 8.1 (95% confidence interval: 5.7–11.5), which was significantly higher than the corresponding HR among boys (3.9 [95% confidence interval: 3.2–4.9]).

CONCLUSIONS: The risk for RD is significantly greater among children with ADHD compared with those without ADHD. Among children with ADHD, the risk for RD is the same for boys and girls. However, among children without ADHD, boys are more at risk for RD than girls. Among girls, the magnitude of increased risk for RD associated with ADHD is nearly twice that among boys, because girls without ADHD are less likely to have RD than boys without ADHD. *Pediatrics* 2010;126:e788–e795

Attention-deficit/hyperactivity disorder (ADHD), characterized by developmentally inappropriate and disabling levels of inattentiveness and/or hyperactivity/impulsivity, is often accompanied by associated learning problems.¹ It is not surprising that children with ADHD have difficulties in learning, because their inattentive and/or hyperactive/impulsive behaviors in school may impede their academic productivity. Although it has also been thought that specific learning disabilities, such as reading disability (RD) or dyslexia, occur more commonly in children with ADHD than in those without it,² results of previous studies that focused on the comorbidity between ADHD and learning disabilities (including RD) have been inconsistent.¹⁻⁷ This is primarily a result of inconsistency in the diagnostic threshold for RD among studies.^{1,8} Furthermore, because most of these studies involved clinically referred samples, results may have been influenced by referral bias. For instance, if children who have both ADHD and RD are more likely to be referred because of increased difficulties in school compared with those with only ADHD or RD, the observed comorbid rate between ADHD and RD might be overestimated. Because disruptive behaviors of boys with ADHD might lead to increased referrals compared with girls with less disruptive behavior, an overestimation of the male/female ratio may occur in studies of clinic-referred samples. However, Willcutt et al⁹⁻¹¹ conducted a series of twin studies regarding comorbidity between ADHD and RD and concluded that these disorders co-occur more frequently than expected by chance even in community samples. In these studies, teachers and parents rated ADHD symptoms. However, as the authors acknowledged,^{2,9} the possibility remains that parents or teachers may be more likely to endorse ADHD symptoms when they know that the child is experiencing difficulty in

learning to read. Furthermore, the characteristics of the twin-study design do not always reflect those of the general population.

To address limitations of previous studies, we evaluated the incidence rate of research-identified RD among children with and without research-identified ADHD in children from a population-based birth cohort, with explicit assessment of gender differences in the comorbidity between these 2 disorders.

METHODS

Study Setting and Data Source

Rochester, Minnesota, is 90 miles southeast of Minneapolis-St Paul, the closest major urban center. In 1990, when subjects in this birth cohort were school-aged children, there were 70 745 residents who were 96% white, fairly young (72% were ≤ 45 years old), and primarily middle class. The demographic characteristics of Olmsted County (Rochester standard metropolitan statistical area) residents resembled those of the white US population during the time frame relevant to this study.¹²

The capacity for population-based epidemiologic research on RD and ADHD in Rochester is the result of a unique set of circumstances. First, Rochester is relatively isolated in southeastern Minnesota; as a result, virtually all medical care is provided locally by the Mayo Clinic, the Olmsted Medical Center, and their 3 affiliated hospitals. Through the Rochester Epidemiology Project, all diagnoses and surgical procedures recorded at the Rochester medical facilities are indexed for automated retrieval.^{12,13} The medical record includes a detailed history of all encounters in the community, including psychiatry, psychology reports, and psychological test results from birth until patients no longer reside in the community. Second,

through a contractual research agreement, all 41 public, parochial, and private schools in Minnesota Independent School District 535 (the school system for the city of Rochester) gave us permission to access their richly documented cumulative educational records for every child from our birth cohort. These cumulative school records are permanently maintained for all children who have attended public, private, or home school in the district. Third, under a separate research agreement we also obtained permission to access the resources of the privately owned Reading Center/Dyslexia Institute of Minnesota (the only private tutoring agency in existence in the community during the school years of our birth cohort members), as well as of the only private community psychiatric practice in the area. The study was approved by the institutional review boards of the Mayo Clinic and the Olmsted Medical Center.

Birth Cohort

Our birth cohort consisted of all children born between January 1, 1976, and December 31, 1982, to mothers residing in the townships comprising Minnesota Independent School District 535 ($n = 8458$). The target population consisted of 5718 children (2956 boys and 2762 girls) who still lived in Rochester at or after the age of 5 years^{14,15} who were followed retrospectively from birth until the initial occurrence of either death, emigration, or high school graduation. The steps and resources used for identification and follow-up of this birth cohort, and analysis of potential influence of migration bias, have already been reported.¹⁶

RD and ADHD Incidence Cases: Identification and Case Definition

Our strategy for identifying all RD and ADHD incidence cases consisted of several steps, used multiple sources of information, and relied on recorded his-

tory of symptoms, individual test results, and treatment, available for all members of the birth cohort. The details of the case-identification procedures were previously described in detail.^{14,15} In short, several steps were used to narrow the pool of potential RD and ADHD incidence cases, starting with cumulative school records of each child in the birth cohort ($n = 5718$). School records were searched for any indication of concerns about learning and behavior, and 1961 children had those concerns observed by teachers, parents, school psychologists, physicians, social workers, and school nurses. Additional work on these 1961 children consisted of abstracting data from the school, medical records, and the records from 2 other private facilities (described above).

The following data were abstracted: all individually administered academic-achievement and cognitive-ability test results and detailed information related to behavioral problems (symptoms, clinical diagnoses, results from teacher/parent questionnaires, medication treatment). Identification of RD incidence cases consisted of applying 3 psychometric criteria. Specifically, for each child designated with learning/behavioral concern, all reading-achievement and IQ test scores were used to form pairs of cognitive-ability and reading-performance measures within each calendar year. The details

of the 3 formulas were published previously.¹⁴ In short, in each of the formulas, x represents the study subject's IQ score, and y represents the standard score from the reading-achievement test. Children classified as having RD by the Regression Formula-Minnesota¹⁷ ($y < 17.40 + 0.62x$) had standard scores in reading achievement that were >1.75 SD below their predicted standard score from an individually administered measure of cognitive ability (IQ). In the discrepancy-formula approach, differences between age-based standard scores of measures of individually administered intelligence and reading achievement varied according to grade (ie, $x - y \geq 15, 19$, or 23 points, for kindergarten through 3rd, 4th–6th, or 7th–12th grade, respectively). Finally, the low-achievement formula ($x \geq 80$ and $y \leq 90$) represents an alternative method for identifying RDs.^{18–20}

Identification of ADHD incidence cases consisted of applying research criteria to the 1961 children (34% of the birth cohort) from our birth cohort who had any recorded behavioral or learning concerns. Subjects were defined as having research-identified ADHD if their school and/or medical records included various combinations of the following 3 different categories of information: (1) meets *Diagnostic and Statistical Manual of Mental Disorders, Fourth Edition* (DSM-IV) criteria for ADHD; (2) positive ADHD questionnaire results²¹; and (3)

clinical diagnosis of ADHD (with or without specific subtype) was documented. Details of information regarding those criteria as well as the identification process of subjects with ADHD were described elsewhere.¹⁵ A total of 379 ADHD cases were identified (Table 1).

Subjects Without ADHD

All the members of the population-based birth cohort who still lived in Rochester at or after the age of 5 years, who were not identified as having ADHD, and who did not have severe intellectual disability were designated as subjects without ADHD.

Statistical Analysis

The Kaplan-Meier method was used to calculate the probability of not meeting the RD criteria from birth to 19 years of age, taking into account the varying duration of each subject's follow-up. In the absence of competing risks, the cumulative incidence of RD was calculated as 1 minus the Kaplan-Meier probabilities and plotted versus age (Figs 1 and 2). Children were classified as having RD by any of the 3 formulas on the date when they initially met the criteria for 1 of the 3 RD formulas. Children who did not meet research criteria for RD were censored on the initial occurrence of migration from the community, death, or last follow-up date or at 19 years of age.

The Cox proportional hazards model was applied to obtain hazard ratios (HRs) and corresponding 95% confidence intervals (CIs). In each Cox model, the incidence of RD was regarded as the outcome variable, whereas ADHD (cases versus non-cases) and gender (boys versus girls) were regarded as explanatory variables. The assumptions of proportional hazards were assessed by graphical methods (plotting the scaled Schoenfeld residuals versus ranked time) and by introducing a time-

TABLE 1 Research Criteria for ADHD Case Definition

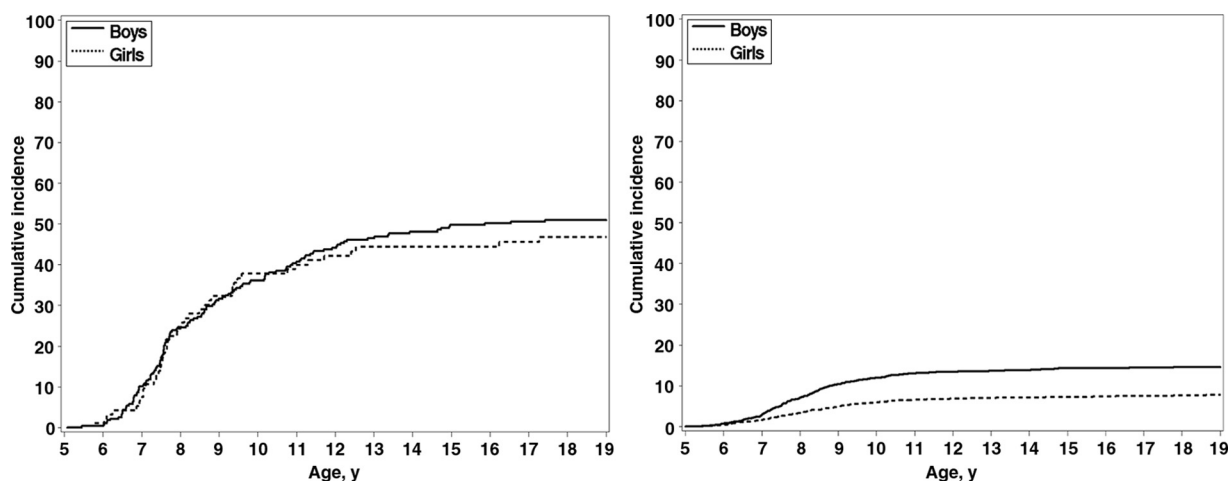
ADHD Cases	Meets DSM-IV Research Criteria for ADHD ^a	ADHD Questionnaire Results	Clinical Diagnosis of ADHD ^b	Number of Subjects
Research identified	+	+	+	170
ADHD cases	+	+	–	41
	+	–	+	17
	–	+	+	122
	–	–	+	29
				228
				151
				379

Pluses and minuses indicate the presence or absence, respectively, of a given criterion. DSM-IV indicates *Diagnostic and Statistical Manual of Mental Disorders, Fourth Edition*.

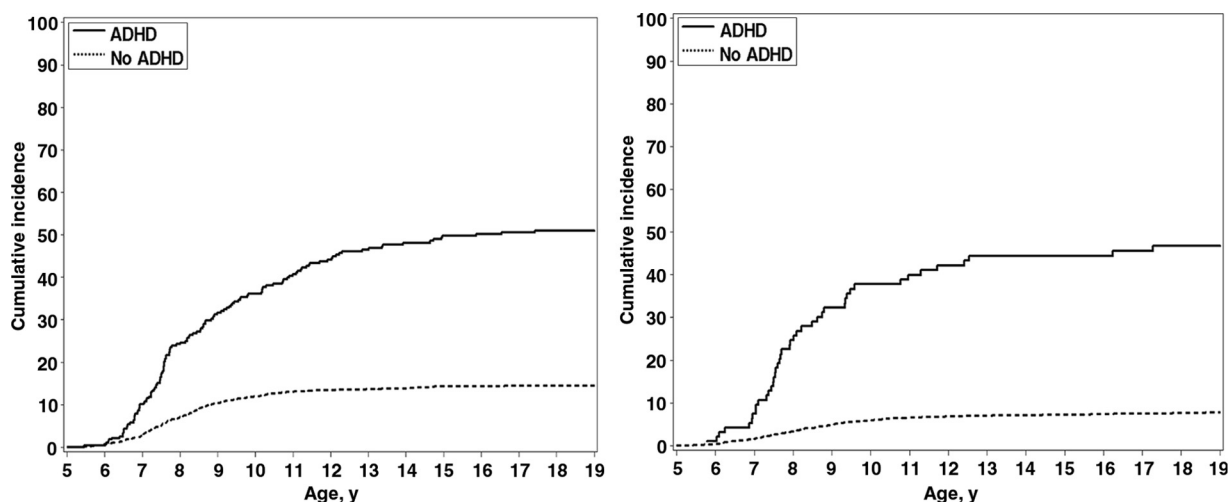
^a All DSM-IV criteria were met, only age criterion was not used.

^b Clinical diagnosis of ADHD (with or without subtype) was recorded in the medical record.

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**FIGURE 1**

Cumulative incidence rates of RD identified by any of 3 formulas for boys and girls, shown separately according to ADHD status. Left, subjects with ADHD; right, subjects without ADHD.

**FIGURE 2**

Cumulative incidence rates of RD identified by any of 3 formulas for children with and without ADHD, shown separately according to gender. Left, boys; right, girls.

dependent coefficient in the Cox models. Both unadjusted and adjusted HRs were calculated. In the latter case, children's race (white versus nonwhite), mothers' educational level, and age at birth of the child were found to be significantly different between subjects with ADHD and those without ADHD (Table 2) and considered to be possibly confounding for the incidence of RD. Therefore, these factors were included in the model. Fathers' education level was not included in the model because of the number of missing values, which may not have been missing at random.

P values (2-sided) of $<.05$ were considered statistically significant.

RESULTS

Among the 5718 subjects in the birth cohort, 19 subjects with severe intellectual disability were excluded, which left 5699 subjects for analysis. A total of 379 children fulfilled the research criteria for ADHD at a mean age of 10.4 years (SD: 3.6; median: 9.8).

Demographic and perinatal factors obtained from the birth certificate of children with and without ADHD in the birth

cohort are shown in Table 2. Subjects with ADHD were significantly more likely to be male ($P < .001$), to be white ($P = .036$), to have parents with fewer years of education ($P = .001$ for fathers and $P = .002$ for mothers), and to have a younger mother at birth ($P = .004$) compared with subjects without ADHD. There were no statistically significant differences regarding perinatal factors between those with and without ADHD.

The cumulative incidence of RD identified by any of the 3 formulas in boys

TABLE 2 Demographic and Perinatal Factors Between Children With and Without ADHD

	Children With ADHD (N = 379)	Children Without ADHD (N = 5320)	P
Child factors			
Boys, n (%)	284 (74.9)	2666 (50.1)	<.001
White (6 missing), n (%)	376 (99.2)	5182 (97.5)	.036
Birth weight < 2500 g (11 missing), n (%)	16 (4.2)	233 (4.4)	.88
Paternal/maternal factors at birth			
Father's education (786 missing), n (%)			.001
<12 y	22 (6.9)	208 (4.5)	
12 y	121 (38.1)	1415 (30.8)	
13–15 y	75 (23.6)	1053 (22.9)	
≥16 y	100 (31.5)	1919 (41.8)	
Mother's education (494 missing), n (%)			.002
<12 y	29 (8.4)	315 (6.5)	
12 y	138 (40.1)	1662 (34.2)	
13–15 y	116 (33.7)	1594 (32.8)	
≥16 y	61 (17.7)	1290 (26.5)	
Father's age (291 missing), mean (SD), y	28.6 (5.5)	28.0 (5.3)	.39
Mother's age, mean (SD), y	25.9 (4.8)	26.6 (4.7)	.004
Father white (215 missing), n (%)	360 (99.5)	5022 (98.1)	.057
Mother white (6 missing), n (%)	377 (99.5)	5203 (97.9)	.035
Marital status (1 missing), n (%)			.12
Married	345 (91.0)	4953 (93.1)	
Not married	34 (9.0)	366 (6.9)	
Pregnancy/labor/delivery factors			
Pregnancy complications, n (%)	37 (9.8)	432 (8.1)	.26
Labor/delivery complications, n (%)	135 (35.6)	1945 (36.7)	.71
Congenital anomalies, n (%)	5 (1.3)	35 (0.7)	.14

Computerized birth certificate information (continuous and dichotomous variables) for all children in the birth cohort were obtained from the Minnesota Department of Health.

and girls and the HR for boys versus girls, listed separately for those with and without ADHD, are listed in Table 3. In the ADHD group, the cumulative incidence of RD was similar between boys and girls, whereas among children without ADHD, boys showed an approximately twofold cumulative incidence of RD compared with girls (Fig 1).

Table 4 and Fig 2 show the risk of RD incidence associated with ADHD separately according to gender. ADHD was

significantly associated with an increased risk of RD in both genders. However, the risk of RD associated with ADHD was significantly higher in girls than in boys (approximately twice as high as that in boys), as shown by the significant ADHD × gender interaction on the incidence of RD ($P = .001$ in the adjusted model). For children with ADHD, the cumulative incidences of RD by 19 years of age, separately for the 3 RD formulas, ranged from 21.1% to

43.2% for boys and from 19.5% to 43.4% for girls.

DISCUSSION

Population-based, nonreferred samples of boys and girls with ADHD and RD are of critical importance to increase our understanding of the natural history of the comorbidity between ADHD and RD. The core of this epidemiologic study is the population-based sample of carefully defined, research-identified ADHD and RD incidence cases.

Our primary findings are:

1. The cumulative incidence of RD is (a) higher in children with ADHD than in those without ADHD in both genders, and (b) there is a significant ADHD × gender interaction on the incidence of RD. Thus, the risk of RD associated with ADHD is higher in girls than in boys.
2. Among children with ADHD, both genders show similar cumulative incidence rates of RD, whereas the cumulative incidence rates of RD are higher in boys than in girls among those without ADHD.

In previous studies that used clinic-referred samples, the diversity of definitions used to diagnose RD has resulted in inconsistent reports of the comorbidity between ADHD and RD.¹ We used 3 formulas for determining RD incidence. The cumulative incidence rate of RD by 19 years of age in

TABLE 3 Risk of RD Associated With Gender, Listed Separately According to ADHD Status

	n	No. of RD Cases by Age 19 y	Cumulative Incidence by Age 19 y, % (95% CI)	Unadjusted HR (95% CI); P^a	Adjusted HR (95% CI); P^b
ADHD					
Boys	284	137	51.0 (44.5–56.6)	1.10 (0.78–1.55); .58	1.04 (0.73–1.50); .83
Girls	95	43	46.7 (35.5–56.0)	1.00	1.00
Non-ADHD					
Boys	2666	332	14.5 (13.0–15.9)	1.97 (1.64–2.36); <.001	2.03 (1.67–2.45); <.001
Girls	2654	176	7.7 (6.6–8.8)	1.00	1.00

RD was as determined by any of the 3 formulas. HR indicates HR for boys versus girls; CI, confidence interval.

^a There is a statistically significant difference between the HR (for boys versus girls) among children with ADHD and that among children without ADHD ($P = .003$ in the unadjusted model and $P = .001$ in the adjusted model).

^b Adjusted for child's race (white versus nonwhite), mother's educational level (4 categories), and mother's age at birth of the child.

TABLE 4 Risk of RD Associated With ADHD, Listed Separately According to Gender

	<i>n</i>	No. of RD Cases by Age 19 y	Cumulative Incidence by Age 19 y, % (95% CI)	Unadjusted HR (95% CI); <i>P</i> ^a	Adjusted HR (95% CI); <i>P</i> ^b
Boys					
ADHD	284	137	51.0 (44.5–56.6)	4.27 (3.50–5.21); <.001	3.94 (3.19–4.85); <.001
Non-ADHD	2666	332	14.5 (13.0–15.9)	1.00	1.00
Girls					
ADHD	95	43	46.7 (35.5–56.0)	7.73 (5.54–10.80); <.001	8.13 (5.73–11.54); <.001
Non-ADHD	2654	176	7.7 (6.6–8.8)	1.00	1.00

RD was as determined by any of the 3 formulas. HR indicates HR for ADHD versus non-ADHD; CI, confidence interval.

^a There is a statistically significant difference between the HR (for ADHD versus non-ADHD) among boys and that among girls (*P* = .003 in the unadjusted model and *P* = .001 in the adjusted model).

^b Adjusted for child's race (white versus nonwhite), mother's educational level (4 categories), and mother's age at birth of the child.

children with ADHD identified by each of the 3 formulas in our population-based birth cohort fell in the range of 20% to 43% for both genders, which is fairly consistent with that of previous studies of clinical samples selected for ADHD^{1,2} and those by Willcutt et al.^{9,11} In addition, our population-based study revealed an overall ~50% cumulative incidence rate of RD (identified by any of the 3 formulas) in children with ADHD for both genders. Thus, the comorbidity between ADHD and RD in our community sample was as high as previously reported in clinical samples.

Although the cumulative incidence of RD in boys was twice as high as in girls among children without ADHD, the similar incidence rates of RD between boys and girls among children with ADHD suggests that ADHD is more strongly associated with RD in girls. This means that both boys and girls with ADHD are at high risk to be affected by RD. The relative preponderance of ADHD subtypes has been reported to vary greatly on the basis of gender.^{22–25} Furthermore, the comorbidity of ADHD and RD has been reported to vary according to ADHD subtype.^{10,11,26} Willcutt et al¹⁰ found that the inattentive symptoms of ADHD were more strongly associated with RD than the hyperactive/impulsive symptoms in a community sample. However, they also implied that the relationship between RD and ADHD subtypes are likely to be complex and, therefore, warrant

additional research.⁹ Moreover, results of recent studies of ADHD subtypes have suggested a lack of stability of subtypes over time and, thus, question their validity as categorical diagnoses.^{27–29} The retrospective nature of this study and changes in the terms of ADHD subtype over time prevented us from obtaining precise information about ADHD subtype for our subjects, and we are therefore unable to address the issue of the influence of ADHD subtype on the extent of comorbidity between ADHD and RD.

Recently, both ADHD and RD were found to be strongly influenced by genetic factors.^{11,26,30} Results of candidate gene and linkage studies of RD and ADHD have suggested the presence of common genes that contribute to both disorders.² There is no additional evidence to support that the 2 disorders are transmitted independently in families.^{31,32} These findings suggest that both RD and ADHD are disorders based, at least in part, on common genetic etiologic factors that increase susceptibility to the comorbidity of both disorders.

Several limitations should be considered when interpreting our results. First, it should be noted that it was not possible for us to determine if symptoms of ADHD preceded those of RD. Because this investigation was a retrospective cohort study in which the relevant information regarding diag-

noses depended on retrospective review of medical and school records, the precise age of onset of both ADHD and RD among children in our sample was not possible to determine; rather, we were able to determine the age at which sufficient information had been documented in the medical and school records to fulfill research criteria for RD and ADHD. It is also possible that some RD or ADHD cases remained unidentified because no screening measures were performed on every child in our birth cohort. However, it should be emphasized that for every child in the birth cohort a systematic, multi-staged, multiresource process was implemented to identify 1961 children (34% of the birth cohort) who had learning-behavior concerns. In addition, we uniformly applied clearly defined research criteria for the identification of all incident cases of ADHD and RD regardless of gender.^{14,15} Finally, at the time of the study, Rochester was primarily a white, middle-class community, so inferences to other populations or settings may be limited. However, Rochester has no access-to-medical-care problems and a homogeneous population (95% white), which thereby minimizes the confounding effect of ethnicity and race on the study questions.

CONCLUSIONS

The results of this population-based birth cohort study indicate that the

incidence of RD is significantly higher in children with ADHD than in those without ADHD in both genders. However, the risk of RD associated with ADHD is significantly higher in girls than in boys, which implies that ADHD is more strongly associated with RD in girls than in boys. In the future, epidemiologic and genetic studies with larger population-based prospective designs and more clearly delineated subtypes of ADHD should lead to a more thorough understanding of the gender differences in the comorbidity between ADHD and RD. Although the American

Academy of Pediatrics clinical practice guideline on the diagnosis and evaluation of children with ADHD does not specifically recommend psychoeducational testing for every child with ADHD,³³ our findings clearly demonstrate that it is essential for clinicians to assess all children with ADHD for the presence of comorbid RD.

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